CIE

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TITLE: **SarthiX**

SCHOOL: Chitkara University, Rajpura, Punjab

DETAIL OF THE STUDENT/MENTOR WHO UPLOADED THE COPYRIGHT ON GOVT PORTAL NEED TO SHARE LOGIN AND PASSWORD WITH MENTORS

Name:

Roll No/ Employee Code:

All MEMBERS DETAILS INCLUDING SUPERVISOR

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ABSTRACT

1.1 Problem Background

On paper, logistics looks smooth. But step inside—it's messy. Drivers don't have a fixed system. They rely on contacts. Random calls. WhatsApp forwards. Sometimes it works. Sometimes not. Income keeps fluctuating. No stability. Shippers face their own issues. They can't compare bids properly. No structured pricing. No transparency. Just guesswork most of the time. And then compliance hits. Permits expire quietly. Documents get outdated. Suddenly fines. Delays. Chaos. Yes, platforms exist. But they only connect. That's it. No depth. No proper system. So yeah. Still broken in many ways.

1.2 Objective of the Project

SarthiX comes in here. Not too complex. Just solving real problems. A place where drivers and shippers interact openly. Transparent bidding. No hidden stuff. Also, permits don't get ignored anymore. System reminds you. Automatically. End goal? Better efficiency. Less confusion. Smarter choices.

1.3 Proposed Solution / System Overview

User logs in. Chooses role. Driver or shipper. Simple start. Driver sees dashboard. Active bids. Earnings. Trips. Ratings. Everything visible. Can place bids. Can update them. But only till timer runs. After that—locked. Shipper creates shipment. Adds details. Waits for bids. Compares. Selects. Tracks shipment live. System handles all this quietly—authentication, dashboards, bidding flow, shipment tracking, permit alerts. Feels smooth. Most of the time.

1.4 Key Features or Functionality

Real-time bidding. That's the heart. Minimum bid control avoids unfair pricing. Dashboards show useful stats. Not too cluttered. Shipment tracking keeps both sides updated. No constant calls needed. Permit module does its job silently. Stores data. Sends reminders. Prevents last-minute panic. Small features. But impactful.

1.5 Technology / Method Used

Frontend built using React with Vite. Fast loading. Clean UI.

Clerk handles authentication. Secure enough for users.

Supabase manages backend and database. Everything synced.

Client-server architecture keeps things structured. Scalable as well.

1.6 Expected Outcome & Benefits

Things become clearer. That's the main shift. Drivers understand earnings better. No blind guessing. Shippers make informed decisions. Better deals. Compliance issues reduce. Fewer penalties. Less stress. Overall flow becomes smoother. Not perfect. But much better.

1.7 Original Contribution Statement

SarthiX combines two worlds. Bidding. And compliance. Most platforms separate them. This one doesn't. That's the difference.

SYSTEM DESIGN

2.1 Architecture

Basic client-server model. Nothing fancy. Frontend talks to backend through APIs. Data flows. Updates happen. Simple structure. Works well.

2.2 Modules

- Authentication — login, signup
- Driver Dashboard — driver's view
- Shipper Dashboard — shipper's control
- Bidding Management — core logic
- Shipment Tracking — real-time updates
- Permit Management — compliance handling

2.3 Data Flow

Level 0: User interacts with system → processes → database

Level 1: Authentication → Dashboard → Bidding → Shipment → Permit

2.4 Use Cases

Actors:

- Driver
- Shipper

Use cases:

- Login/Register
- Place Bid
- Create Shipment
- Track Shipment
- Manage Permits

2.5 ER (Entity Relationship)

Entities:

- User
- Shipment
- Bid
- Permit Relationships map out how these entities interact.

IMPLEMENTATION

3.1 Technologies Used

- Frontend: React (Vite)
- Backend: Supabase
- Authentication: Clerk

3.2 Functionalities

- Real-time bidding system
- Shipment listing and tracking
- Dashboard analytics
- Permit expiry notifications

3.3 Workflow

1. User logs in
2. Picks their role
3. Uses dashboard
4. Manages bids, shipments, or permits

RESULTS AND DISCUSSION

System improved things. Clearly. Pricing became more transparent. Manual coordination reduced. Permit compliance improved. Operations felt faster. Less friction. Still room for metrics and graphs. That part can be added later.

CONCLUSION

SarthiX works as a unified solution. Bidding + compliance together. Makes logistics simpler. More structured. Not perfect yet. But strong foundation.

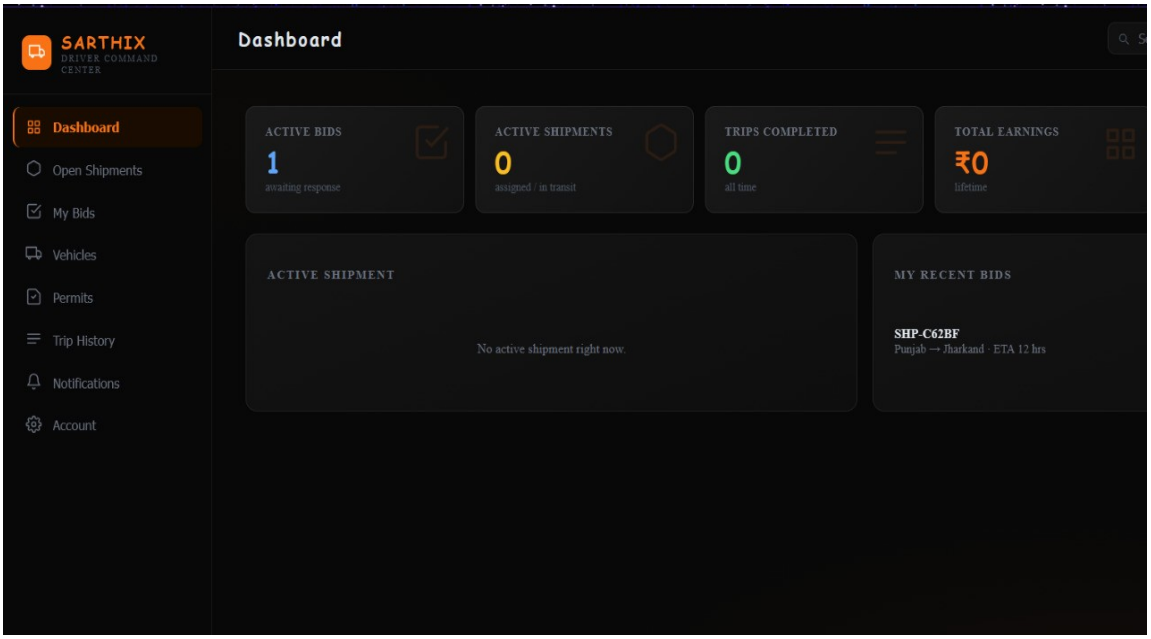
FUTURE SCOPE

- AI-powered bidding tips
- Real-time GPS tracking
- Develop a mobile app
- Integrate payment gateways

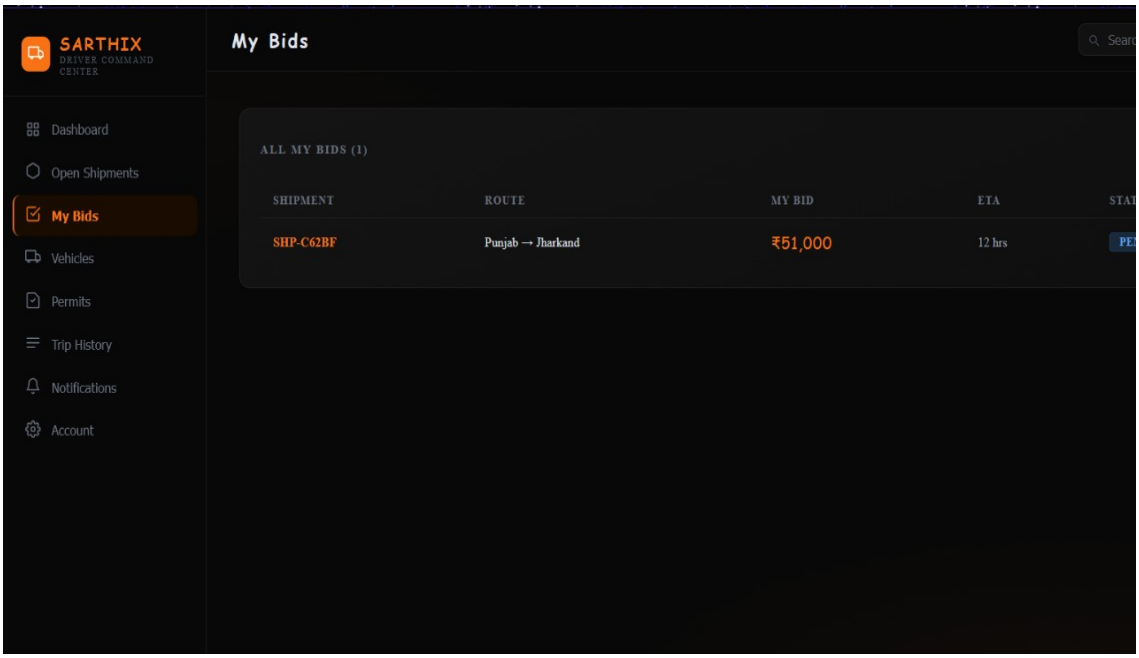
APPENDIX

Screenshots

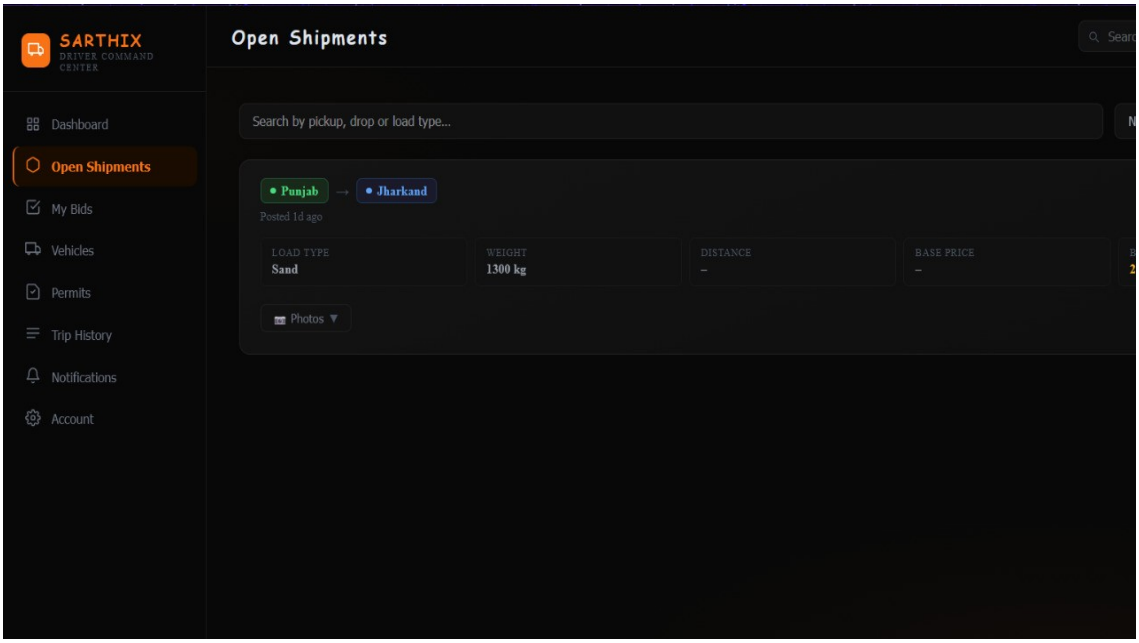
- Figure 1: Dashboard



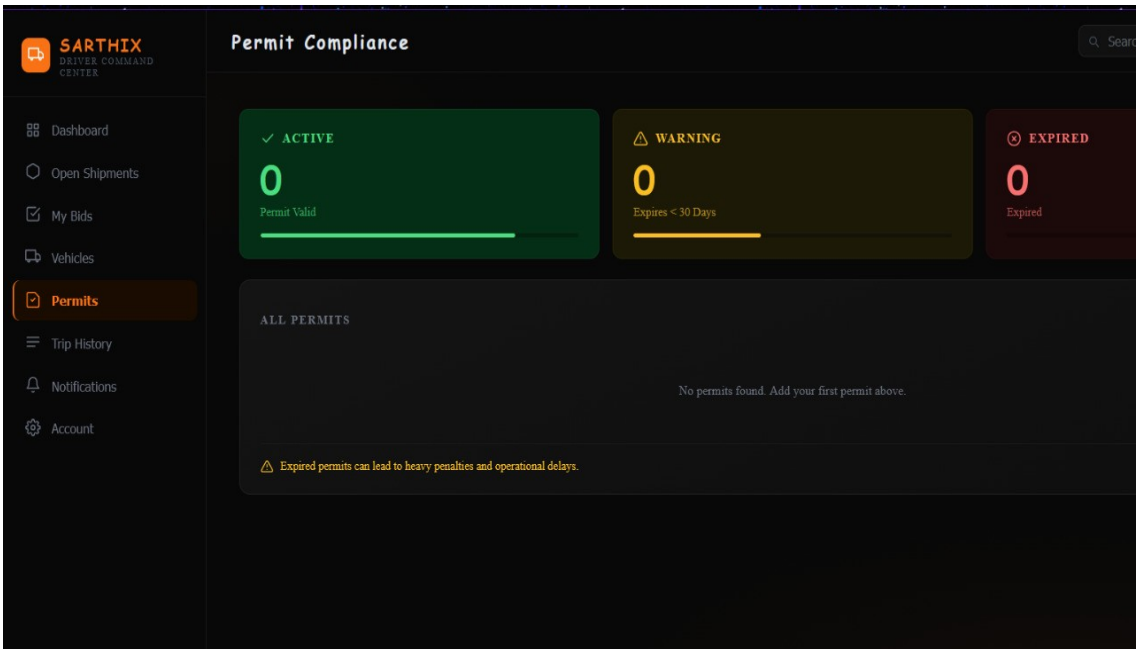
- Figure 2: Bidding Page



- Figure 3: Shipment Page

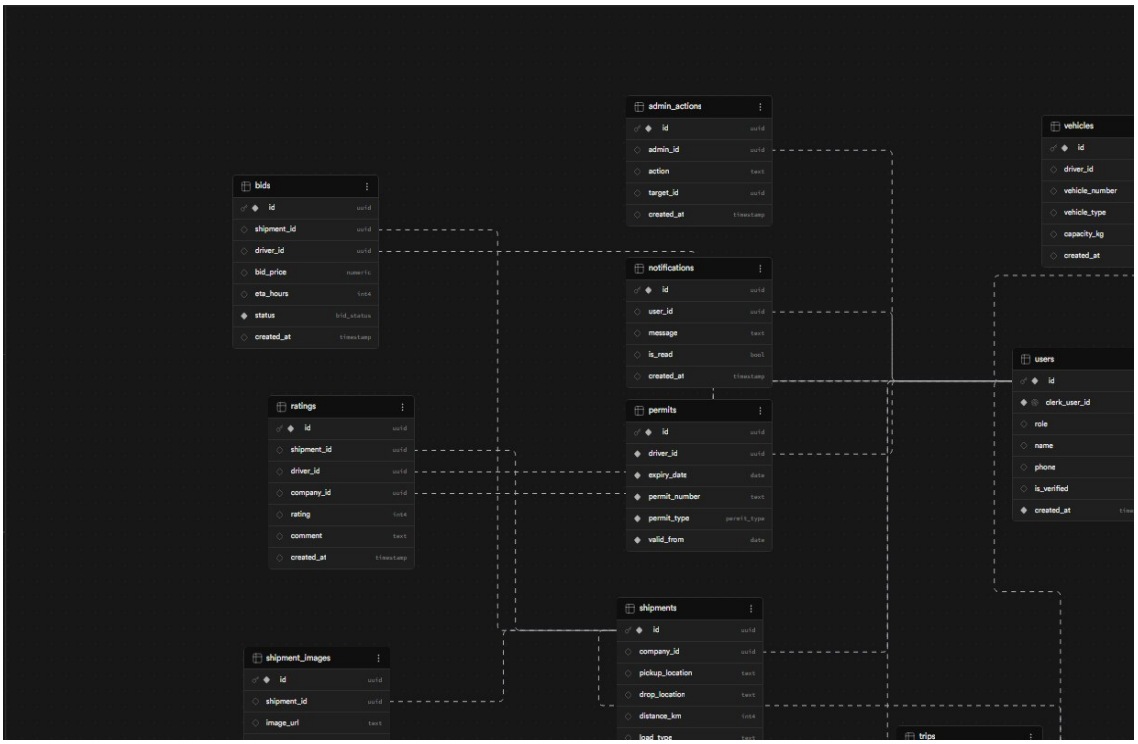


- Figure 4: Permit Page

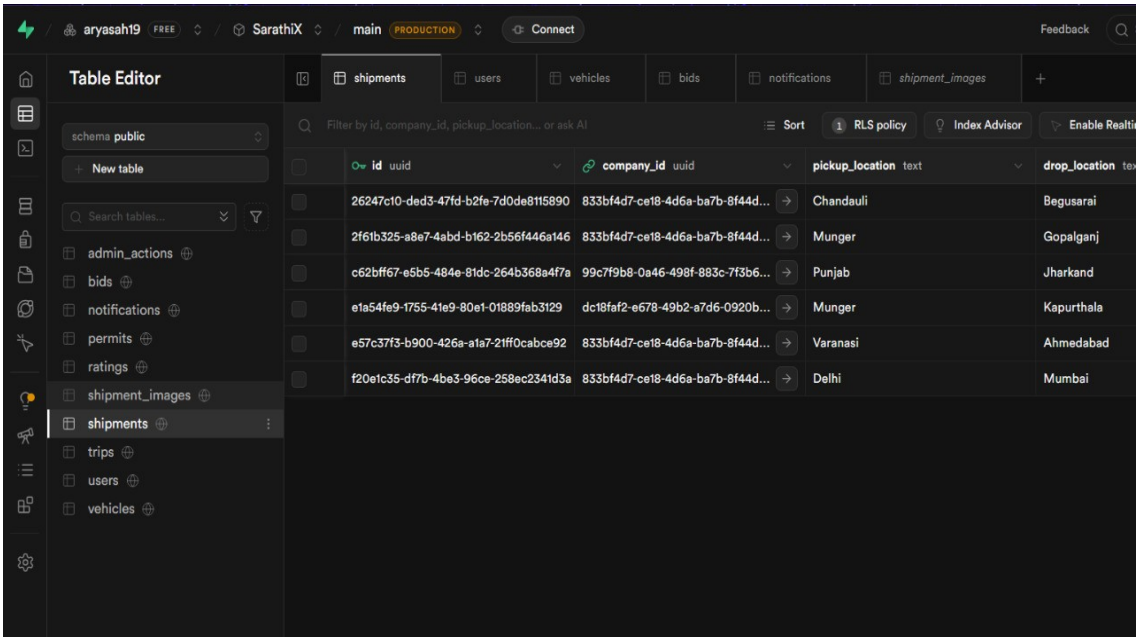


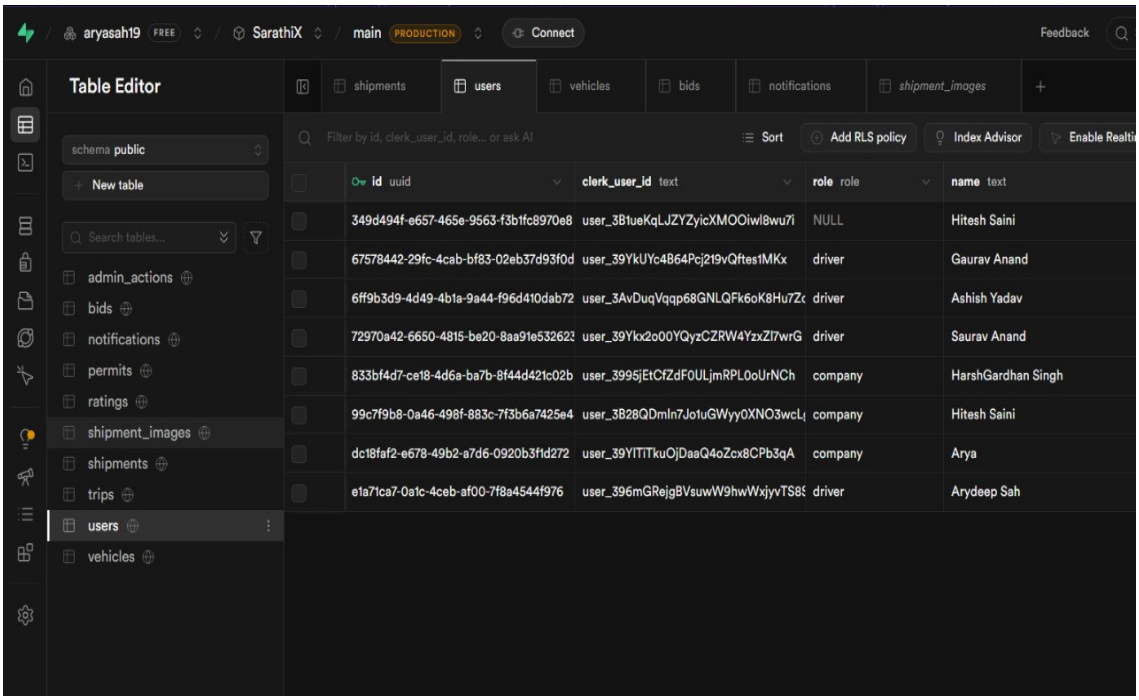
Diagrams

- Data Flow

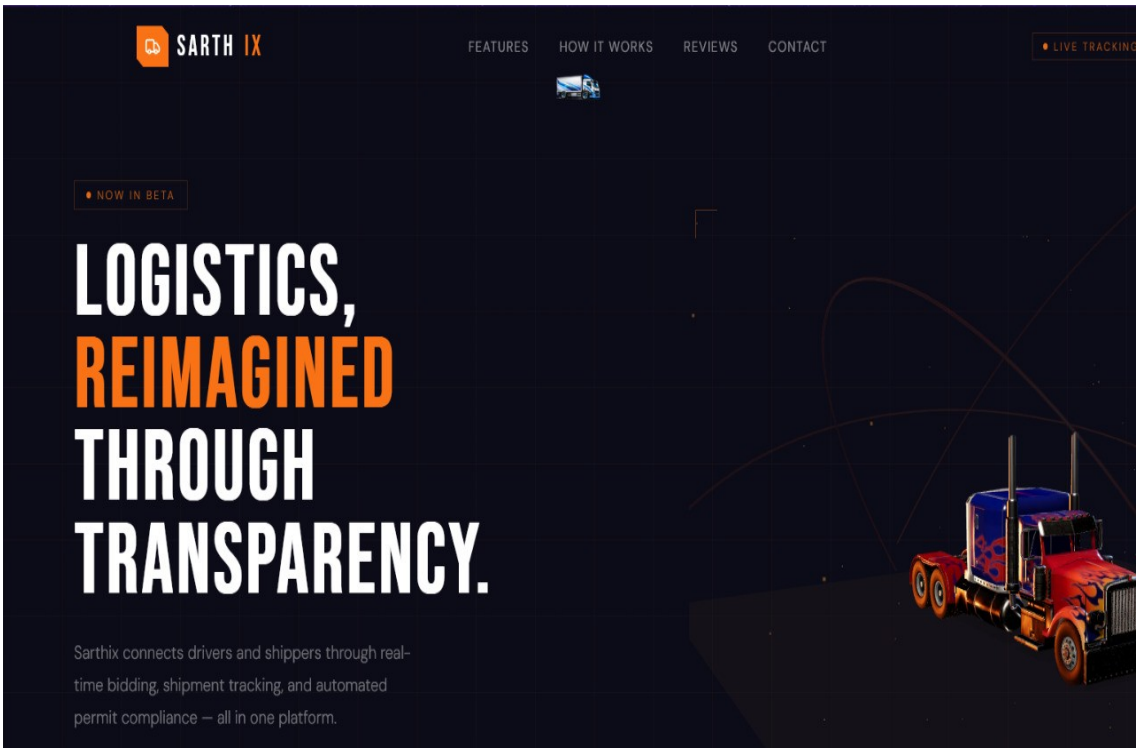


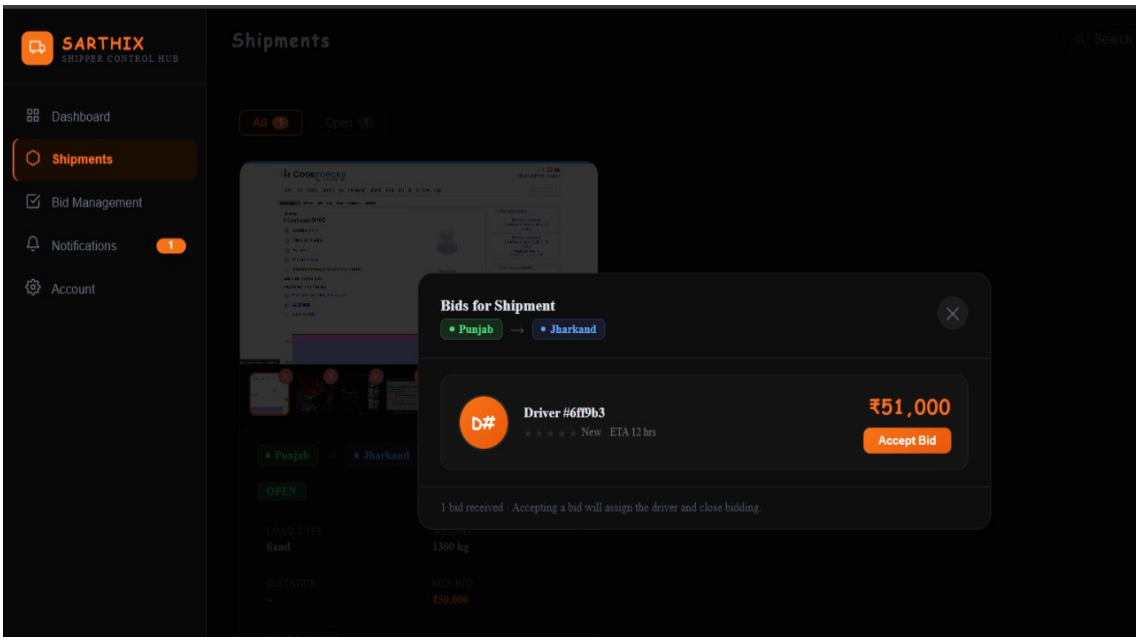
- Database

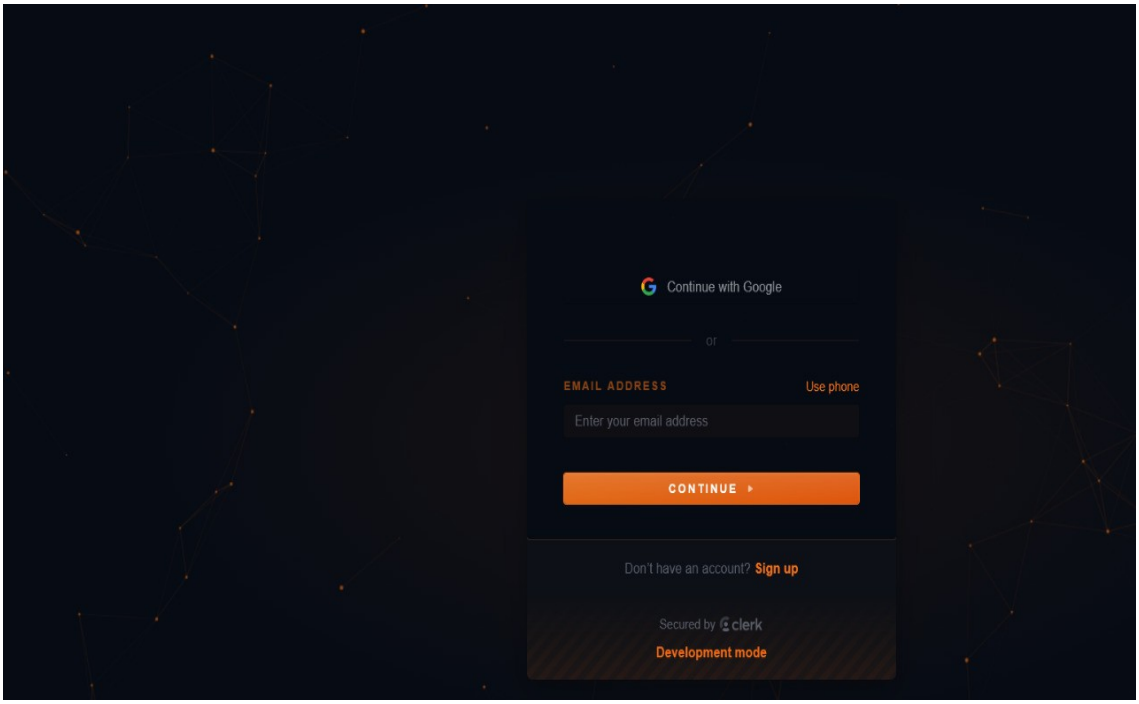




- User Actions




SARTHIX
SHIPPER CONTROL HUB
Dashboard
Shipments
Bid Management
Notifications
1
Account
Shipments
All
Open
Bids for Shipment
Punjab
Jharkand
D#
Driver #6f9b3
New ETA 12 hrs
₹51,000
Accept Bid
1 bid received · Accepting a bid will assign the driver and close bidding.


Continue with Google
or
EMAIL ADDRESS
Use phone
Enter your email address
CONTINUE
Don't have an account? Sign up
Secured by clerk
Development mode